

DIVERSITY OF LEAF ASSOCIATED TRENTEPOHLIAN ALGAL BIOFILMS IN SOUTHERN WESTERN GHATS

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AUTHORS' CONTRIBUTIONS

This work was carried out in collaboration between all authors. Author BTT designed the study and performed the statistical analysis and author MVB wrote the first draft of the manuscript and wrote the protocol and author VPT managed the analyses of the study. All authors read and approved the final manuscript.

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ABSTRACT

This study assessed the diversity of leaf associated trentepohlian algae in the Southern Western Ghats region of Kerala. Trentepohlian algae are terrestrial micro floras which are still a little investigated area of research in comparison to other floral diversity. Epiphyllous algae from forty one different cryptogamic vascular and phanerogamic vascular plants were examined in two different habitat viz. semi-urban and rural areas of Pathanamthitta District, Kerala, India an important biodiversity zone of Western Ghats. Taxonomic characterization of Trentepohlian algae and their host plants were carried out by systematic field studies and microscopic examinations. The present study identified four genera of algae such as *Phycopeltis*, *Cephaleuros*, *Printzina* and *Trentepohlia*. They appear in different habit such as discoid, prostrate and erect forms. *Aporosa cardiosperma* (Gaertn.) Merr. had maximum algal diversity on its leaves with four trentepohlian algal species; followed by *Xanthophyllum flavescens* Roxb. and *Memecylon umbellatum* Burm.f. with three taxa; followed by *Mangifera indica* L. and *Ficus hispida* L.f. with two taxa. While all the other thirty members including the three cryptogamic vascular plants showed least algal diversity. Nevertheless the three members of the family Malvaceae (*Sida acuta* Burm.f, *Hibiscus rosa-sinensis* L. and *Malva viscus* Fabr.) and three members of the order Zingiberales (*Kaempferia galangal* L., *Maranta aurundinacea* L. and *Costus speciosus* J.Koenig) had never showed any algal association. *Phycopeltis* (71%) was the dominant epiphyllous alga, followed by *Cephaleuros* (36%), *Trentepohlia* (14%) and *Printzina* (3%). The present investigation revealed that phyllosphere in the rural area has more trentepohlian algae.

Keywords: *Cephaleuros*, epiphyllous; *Phycopeltis*; *Printzina*; *Trentepohlia*.

1. INTRODUCTION

Usually algae grow in aquatic habitats; but many of the species occur in wet terrestrial environments [1]. Certain green and blue-green algae are frequently found in terrestrial habitats [2]. Among these green algae, Trentepohlian members are well-known species found in terrestrial environments [3,4]. The order Trentepohliales is sub-aerial green algae of the

class Ulvophyceae, represented by a single family Trentepohliaceae [3]. *Phycopeltis* Millardet, *Cephaleuros* Kunze ex E.M. Fries 1832, *Printzina* R.H. Thompson et Wujek 1992, *Trentepohlia* Martius 1817 and *Stomatochroon* Palm 1934 are the five different genera observed from the order Trentepohliales [3]. Currently 70 to 100 species of algae are described from the family Trentepohliaceae [3,5]. All Trentepohliaceae have prostrate to

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filamentous forms. They often contain pigments of β -carotene and haematochrome causing algae to appear yellow orange in colour at the adult stage [3,6]. Presence of special pigments, absence of pyrenoids in the chloroplast and characteristic reproductive structure-the sporangiate lateral are the unique features of the order Trentepohliales [7].

All Trentepohlialean algae occur either as free living or living in association with lichen [8] or plants. Rindi & Guiry [9] reported that some trentepohlian algae showed a preference for certain substrata. They occur either as epiphytes, endophytes or epizoid. Epiphytes algae grow on bark of the trees, leaves, rock and artificial substrata. Endophytic algae parasitically or symbiotically lives in leaves and epizoid on some animals [4,6, 9]. Some species cause trouble in the different vegetation. *Cephaleuros* species are common algal pathogens on leaves of many vascular plants [10, 11, 12].

The epiphyllous algae not only act as parasites or symbionts but also they are involved in nutrient cycle. Similarly they have many economic and ecological implications. Nevertheless there are a few reports of Trentepohliales from Kerala, India [13]. There has been no thorough work on diversity of epiphyllous algae from Kerala, India. This preliminary study of epiphyllous algae from Pathanamthitta District would be beneficial to evaluate the diversity of Trentepohliales.

2. MATERIALS AND METHODS

Epiphyllous algae were collected from five different sites (each was of about 1-2 acre land) of semi-urban and rural area from Pathanamthitta District (9.2601°N, 76.9643°E) in Kerala. The areas were Adoor, Kaipattoor, Konni, Thannithodu and Ranni. The details are shown in the Table 1.

2.1 Collection, Isolation & Preservation

Leaf samples were collected during June to December 2016 and January to April 2017 for observing the diversity of epiphyllous algae. The samples of leaves were preserved as dried herbarium specimens and also kept in polythene bags for further chemical analysis. Voucher specimens were deposited in the Catholicate College Herbarium (CATH), Pathanamthitta, Kerala, India with accession numbers. The specimens of epiphyllous algae associated with host leaves were also preserved in Formalin-Acetic Acid-Alcohol (FAA) for long term storage and future reference. The permanent slides of Trentepohlian algae were also kept in the Phycotechnology Herbarium (CAPH), Catholicate College, Pathanamthitta, Kerala, India with accession numbers.

2.2 Photomicrography

The thalli of *Phycopeltis*, *Trentepohlia*, and *Printzina* were easily isolated from the samples of host leaves [14] for microscopic observation. While the *Cephaleuros* was observed by taking the anatomy of infected leaves. The morphological traits and microscopic features were observed by using the Olympus LX 400 Trinocular microscope and photographs were taken using BioLinkz Cmos Cam (3.0 m pixels) attached to the microscope. Altogether thirty cells per species were measured in the observation [14].

2.3 Identification

Major characterization and classification of algae were based on the monograph [6] and the database [5]. In addition to that, a systematic key [15] was also referred.

2.4 Diversity Analyses

Statistical analysis of algal diversity was carried out by PAST package software.

3. RESULTS

Altogether five genera of Trentepohliales such as *Phycopeltis*, *Cephaleuros*, *Printzina* and *Trentepohlia* (Fig. 1) were observed from thirty five different plants.

3.1 Host Plants of *Phycopeltis* (Fig.2)

Altogether 6 species of *Phycopeltis* were isolated from leaves of different host plants. The species were *P. aurendinacea* (Montagne) De Toni, *P. epiphyton* Millardet, *P. irregularis* (Schmidle) Wille, *P. aurea* Karsten, *P. vaga* Thompson et Wujek and *P. flabellata* Thompson et Wujek. *Phycopeltis* was the most diverse species in all samples. It was observed in 71% cryptogamic and phanerogamic vascular plants in the study area. (Table 2).

3.1.1 *Phycopeltis aurendinacea* (Montagne) De Toni (Accession number-CAPH 121)

It has disk shaped thalli of 3-6 mm in diameter with laterally appressed dichotomous filaments. The thalli had more or less lobed margin. The thallus consisted of orange colour in the adult stage. It was isolated from the semi-urban and natural forest regions during the three seasons such as southwest, northeast and summer seasons. The host plants were *Memecylon umbellatum* Burm.f. (Family-Melastomaceae,

Table 1. Study sites of *Phycopeltis* spp.

Sl. No	Place	Nature of the site	Latitude	Longitude
1	Adoor, Kerala	Semi- urban	9.1512 ⁰ N	76.7308 ⁰ E
2	Kaipattoor, Kerala	Semi -urban	9.2251 ⁰ N	76.7510 ⁰ E
3	Konni, Kerala	Rural area-Natural Forest	9.2267 ⁰ N	76.8497 ⁰ E
4	Thannithodu, Kerala	Rural area- Natural Forest	9.2644 ⁰ N	76.9355 ⁰ E
5	Ranni, Kerala	Rural area- Natural forest	9.3866 ⁰ N	76.7856 ⁰ E

CATH-12043, Habitat-Thannithodu), *Adiantum caudatum* L. (Family-Adiantaceae, Pteridophyte, CATH-12021, Habitat- Adoor), *Drynaria quercifolia* (L.) Sm. (Family-Polypodiaceae-Pteridophyte, CATH-12017, Habitat-Adoor), *Macaranga peltata* M.Arg. (Family-Euphorbiaceae, CATH-12005, Habitat-Ranni), *Chrysophyllum cainito* L. (Family-Sapotaceae, CATH-12008, Habitat-Kaipattoor), *Pteris vitata* L. (Family-Pteridophyte, CATH -12013, Habitat-Ranni), *Coffea arabica* L. (Family-Rubiaceae, CATH-12015, Habitat-Thannithodu), *Jasminum grandiflorum* L. (Family-Oleaceae, CATH-12016, Habitat-Adoor), *Strychnos nux vomica* L. (Family-Loganiaceae, CATH-12020, Habitat-Konni), *Wattakaka volubilis* (Family-Asclepiadaceae, CATH-12021, Habitat-Adoor), *Annona reticulata* L. (Family-Annonaceae, CATH -12034, Habitat-Thannithodu), *Ixora finlaysoniana* Wall. (Family-Rubiaceae, CATH-12036, Habitat-Konni), *Ficus hispida* L.f. (Family-Moraceae, CATH-12014, Habitat-Adoor), *Cananga odorata* (Lam.) Hook. f. & Thoms (Family-Annonaceae, CATH-12037, Habitat- Thannithodu), *Mangifera indica* L. (Family-Anacardiaceae, CATH-12039, Habitat-Konni), *Aristolochia indica* L. (Family-Aristolochiaceae, CATH-12038, Habitat-Adoor), *Caryota urens* L. (Family-Palmae, CATH-12025, Habitat- Konni) and *Xanthophyllum flavescens* Roxb. (Family-Polygalaceae, CATH-12026, Habitat-Kaipattoor). The species was present in 51% of the host plants.

3.1.2 *Phycopeltis epiphyton* Millardet (Accession number-CAPH 122)

It has disc shaped thallus of 1.5 to 2 mm in diameter. At maturity it becomes greenish-orange coloured. It was usually found at the region of rural area-natural forest and very rarely observed at the semi-urban regions. The species was isolated in South-west and North-east monsoon seasons. The host plants were *Coffea arabica* L. (Family- Rubiaceae, CATH -12062, Habitat-Adoor) and *Cyclea peltata* Arn. (Family- Menispermaceae, CATH -12009, Habitat-Adoor). It was present in 6% of the host plants.

3.1.3 *Phycopeltis irregularis* (Schmidle) Wille (Accession number-CAPH 124)

The species has no regular disc. Filaments of the specimens were branched, wide spreading, and

dichotomy. It was present in all the studied area except the semi-urban area like Kaipattoor and Adoor of Kerala. It was more common in the southwest and northeast monsoon seasons. It was examined from *Bridelia retusa* Spr. (Family-Euphorbiaceae, CATH-12042, Habitat-Thannithodu) and *Aporosa cardiosperma* (Gaertn.) Merr. (Family-Euphorbiaceae, CATH-12041, Habitat- Thannithodu). It was also present in 6% of the host plants.

3.1.4 *Phycopeltis aurea* Karsten (Accession number-CAPH 125)

The thalli of the species were circular in nature. Filaments were radiating, laterally appressed and dichotomous. It was observed in the semi-urban area of the study site. It was present in northeast monsoon and summer seasons. The host plant was *Mimusops elengi* L. (Family-Sapotaceae, CATH-12011, Habitat-Kaipattoor). Only 3% of the host plants had the association with *P. aurea*.

3.1.5 *Phycopeltis flabellata* Thompson and Wujek (Accession number-CAPH 123)

The species has many fan like ramuli appressed to form more or less closed disks. It grows up to 2 mm in diameter. It was found in all the studied semi-urban and rural-natural forest regions during southwest and northeast monsoon seasons. The host plants were *Alpinia calcarata* Rosc. (Family- Zingiberaceae, CATH-12032, Habitat- Kaipattoor), *Mimusops elengi* L. (Family-Sapotaceae, CATH-12063, Habitat-Thannithodu) and *Anacardium occidentale* L. (Family-Anacardiaceae, CATH- 12030, Habitat-Kaipattoor). Only 9% of the host plants made relationship with *P. flabellata*.

3.1.6 *Phycopeltis vaga* Thompson et Wujek (Accession number-CAPH 126)

The thalli were open and observed dichotomizing filaments. It was isolated from the rural- natural forest. It was found in all the seasons except summer. *Myristica beddomei* King. (Family- Myristicaceae, CATH-12050, Habitat- Ranni, Konni and Thannithodu) was the only host of *P.vaga* identified during the study.

3.2 Host Plants of *Cephaleuros* (Fig. 3 A & B)

Altogether 36% of phanerogamic vascular host plants showed the presence of *Cephaleuros*. *C. virescens* and *C. diffusus* were the two species identified.

3.2.1 *Cephaleuros virescens* Kunze (Accession number -CAPH 128)

Transverse sectioning of the infected leaves showed that thalli of algae were sub cuticular or sub epidermal growth through the leaf tissue and caused necrosis of cells. Circular disk of thalii were composed of open filamentous. Setae were rarely found as cylindrical filaments. It was usually found in the semi-urban and rural area of the study sites. It was present in the southwest and northeast monsoon seasons of Kerala. The host leaves were *Lagerstroemia speciosa* L. (Family-Lythraceae, CATH 12003, Habitat-Konni), *Piper betle* L. (Family- Piperaceae, CATH-12027,

Habitat-Adoor), *Ficus hispida* L.f. (Family-Moraceae, CATH- 12014, Habiata-Adoor), *Tabernamontana divaricata* R.Br.ex Roem. & Schult (Family- Apocynaceae, CATH-12031, Habitat-Adoor) *Nephelium lappaceum* L. (Family-Sapindaceae, CATH-12036, Habitat-Thannithodu), *Mangifera indica* L. (Family-Anacardiaceae, CATH-12043, Habitat-Thannithodu), *Strobilanthes heyneanus* Nees (Family-Acanthaceae, CATH-12049, Habitat-Ranni), *Bridelia retusa* Spr. (Family-Euphorbiaceae-CATH-12042, Habitat-Thannithodu), *Alstonia scholaris* R.Br. (Apocynaceae-CATH-12055, Habitat-Ranni), *Piper nigrum* L. (Family-Piperaceae-CATH-12019,Habitat-Kaipattoor), *Aporosa cardiosperma* (Gaertn.) Merr. (Family-Euphorbiaceae, CATH-12041, Habitat- Thannithodu) and *Xanthophyllum flavescens* Roxb. (Family-Polygalaceae, CATH-12026, Habitat- Kaipattoor). The alga was found in 34% of the host plants and made parasitic relationship.

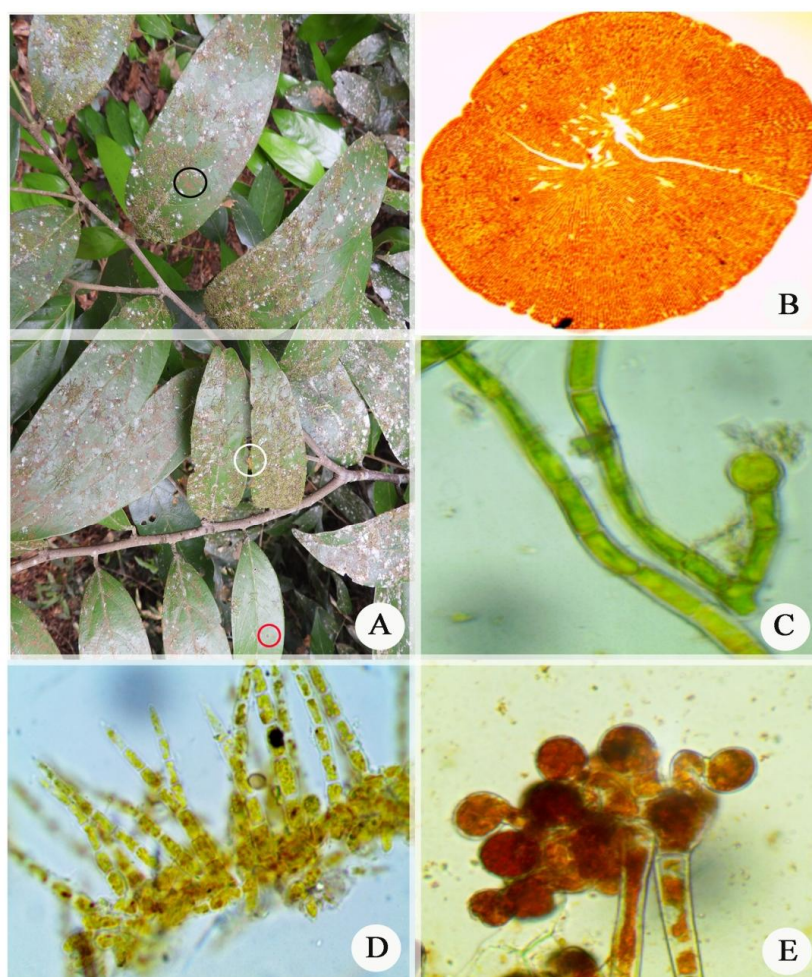


Fig .1.A.*Aporosa cardiosperma* (Host plant)with algal interactions;
B.*Phyceltis aurendinacea* ;C.*Trentepohlia abietina*;D. *Printzina* species;E. *Cephaleuros virescens*

Table 2. Habitat, host plants and distribution of Trentepohlian algae

Sl.No	Host plants	Habitat	Trentepohlian algae			
			<i>Phycopeltis</i>	<i>Cephaleuros</i>	<i>Printzina</i>	<i>Trentepohlia</i>
1	<i>Memecylon umbellatum</i>	Thannithodu	+	-	+	+
2	<i>Adiantum caudatum</i>	Adoor	+	-	-	-
3	<i>Drynaria quercifolia</i>	Adoor	+	-	-	-
4	<i>Macaranga peltata</i>	Ranni	+	-	-	-
5	<i>Chrysophyllum cainito</i>	Ranni	+	-	-	-
6	<i>Pteris vitata</i>	Ranni	+	-	-	-
7	<i>Coffea arabica</i>	Thannithodu & Adoor	+	-	-	-
8	<i>Jasminum grandiflorum</i>	Adoor	+	-	-	-
9	<i>Strychnos nux vomica</i>	Konni	+	-	-	-
10	<i>Vattakakka volubilis</i>	Adoor	+	-	-	-
11	<i>Anona reticulata</i>	Thannithodu	+	-	-	-
12	<i>Ixora finlaysoniana</i>	Konni	+	-	-	-
13	<i>Ficus hispida</i>	Adoor	+	+	-	-
14	<i>Cananga odorata</i>	Thannithodu	+	-	-	-
15	<i>Mangifera indica</i>	Thannithodu & Konni	+	+	-	-
16	<i>Aristolochia indica, L.</i>	Adoor	+	-	-	-
17	<i>Caryota urens, L.</i>	Konni	+	-	-	-
18	<i>Cyclea peltata</i>	Adoor	+	-	-	-
19	<i>Bridelia retusa</i>	Thannithodu	+	-	-	-
20	<i>Aporosa cardiosperma</i>	Thannithodu	+	+	+	+
21	<i>Mimusops elengi</i>	Kaipattoor	+	-	-	-
22	<i>Alpinia calcarata</i>	Kaipattoor	+	-	-	-
23	<i>Anacardium occidentale</i>	Kaipattoor	+	-	-	-
24	<i>Myristica beddomei</i>	Thannithodu, Konni & Ranni	+	-	-	-
25	<i>Lagerstroemia speciosa</i>	Konni	-	+	-	-

Sl.No	Host plants	Habitat	Trentepohlian algae			
			<i>Phycopeltis</i>	<i>Cephaleuros</i>	<i>Printzina</i>	<i>Trentepohlia</i>
26	<i>Piper nigrum</i>	Kaipattoor	-	+	-	-
27	<i>Alstonia scholaris</i>	Ranni	-	+	-	-
28	<i>Strobilanthes heyneanus</i>	Ranni	-	+	-	-
29	<i>Nephelium lappaceum</i>	Thannithodu	-	+	-	-
30	<i>Tabernamontana divaricata</i>	Adoor	-	+	-	-
31	<i>Piper betle</i>	Adoor	-	+	-	-
32	<i>Artocarpus incisus</i>	Konni	-	+	-	-
33	<i>Xanthophyllum flavescens</i>	Kaipattoor	+	+	-	+
34	<i>Artocarpus integrifolia</i>	Adoor	-	-	-	+
35	<i>Vateria indica</i>	Ranni	-	-	-	+
36	<i>Kaempferia galanga</i>	Adoor	-	-	-	-
37	<i>Sida acuta</i>	Konni, Thannithodu, Ranni, Kaipattoor & Adoor	-	-	-	-
38	<i>Hibiscus rosa-sinensis</i>	Konni, Thannithodu, Ranni, Kaipattoor & Adoor	-	-	-	-
39	<i>Maranta aurundinacea</i>	Ranni, Adoor	-	-	-	-
40	<i>Costus speciosus</i>	Konni, Ranni, Adoor & Kaipattoor	-	-	-	-
41	<i>Malva viscus</i>	Adoor	-	-	-	-

Abbreviations: + - Present; - Absent

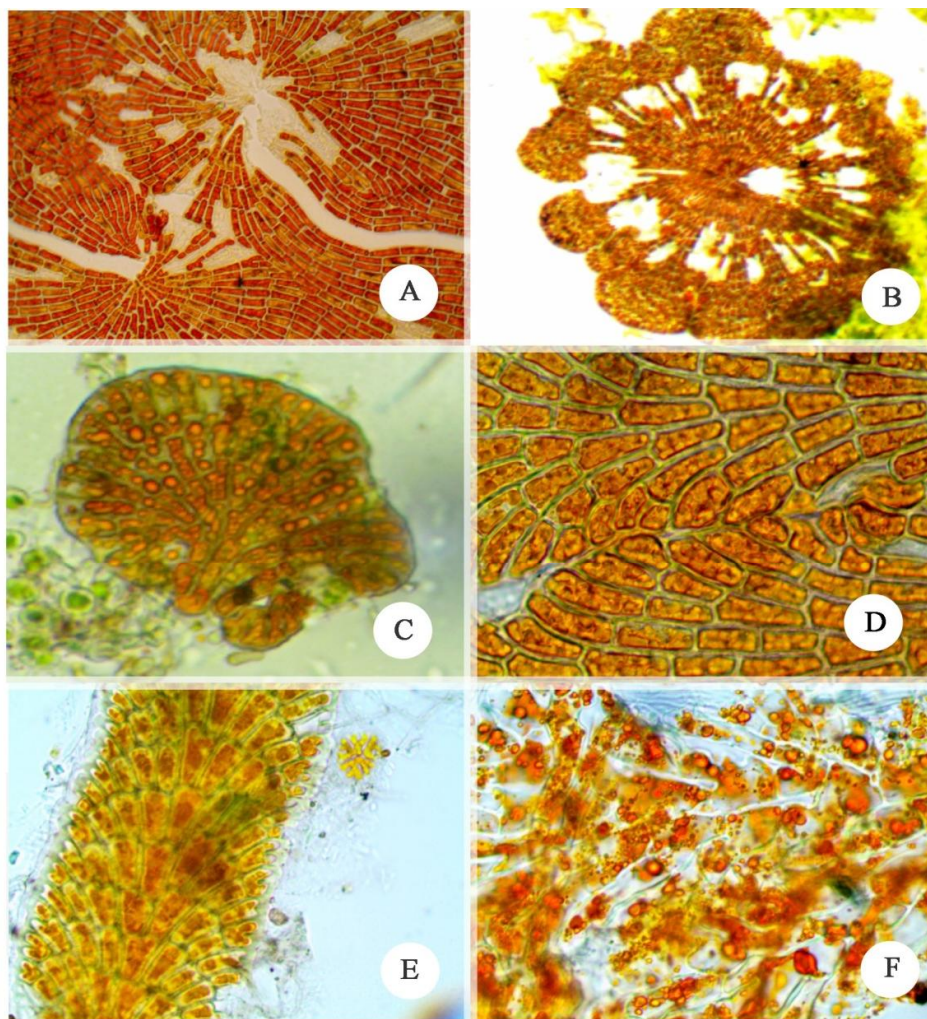


Fig 2. Diversity of *Phycopeltis* species in the study area. A. Thallus of *P. aurendinacea*; B. Thallus of *P. epiphyton*; C. Thallus of *P. aurea*; D. Thallus of *P. flabellata*; E. Thallus of *P. vaga*; F. Thallus of *P. irregularis*

3.2.2 *Cephaleuros diffusus* Thompson & Wujek (Accession Number-CAPH 129)

The thalli were found as spot, filamentous cells were short cylindrical to irregular. It was usually found in the semi-urban and rural area of the study sites. It was usually present in the northeast and monsoon seasons of Kerala. The host leaves were *Artocarpus incisus* (Thunb.) L.f. (Family-Moraceae, CATH-12001, Habitat-Konni). Only 3% of plant exhibited association with *C. diffusus*.

3.3 Host Plant of *Printzina* sp. (Accession Number CAPH 127) (Fig.3 C & D)

Thallus consists of poorly branched, erect pointed axes, arising from an extensive system of prostrate

filaments. It was examined from the rural area of natural forest during the southwest and northeast monsoon seasons. It was examined from the leaves of *Aporosa cardiosperma* (Gaertn.) Merr. (CATH-12041), Habitat- Thannithodu). It was found only in 3% of the plants.

3.4 Host Plants of *Trentepohlia* spp. (*T. abietina* and *T. rigidula* (Accession number CAPH 130 & 131) (Fig. 3 E, F & G)

The plant body was thalloid and composed of branched uniseriate filaments. Filaments were arranged in erect tufts. Sporangia arise from the vegetative cells which were distinguishable due to its large swollen structure. It was profoundly examined from the semi-urban areas during all seasons. It was

isolated from the plants of *Xanthophyllum flavescens* Roxb. (Family- Polygalaceae, CATH-12026, Habitat-Kaipattoor- *T. abietina*), *Artocarpus integrifolia* L. (Family-Moraceae, CATH-12006, Habitat-Adoor) *Vateria indica* L. (Family-Dipterocarpaceae, CATH-12058, Habitat-Ranni), *Aporosa cardiosperma* (Gaertn.) Merr. (Family-Euphorbiaceae, CATH-12041, Habitat- Thannithodu) and *Memecylon umbellatum* Burm.f. (Family-Melastomaceae, CATH-12048, Habitat- Thannithodu). It was isolated from 14% of the host plants.

The statistical analysis (Fig. 4) revealed that, variability of all algae was significant. It was noted

that *Phycopeltis* spp. are abundantly present (Fig. 5) in all the area. *Printzina* sp. was exclusively reported from the rural areas of natural forest regions. Quite interestingly it was observed that the parasitic algae, *Cephaleuros* spp. were detected from the semi-urban areas. *Trentepohlia* sp. was also exclusively present in different semi-urban areas. Maximum diversity of algae was found in rural area especially in natural forest of Thannithodu (Fig. 6).

Aporosa cardiosperma harbours maximum diversity with 4 genera of trentepohlian algae. *Xanthophyllum flavescens* and *Memecylon umbellatum* had 3 types of trentepohlian algae; followed by *Mangifera indica*



Figs 3. Diversity of different Trentepohliales in Pathanamthitta district
A. *Cephaleuros virescens* ;B. *C. diffusus* ; C&D. *Printzina* spp.;E&F. *Trentepohlia rigidula*;
G. *T. abietina*.

and *Ficus hispida* with two taxa (Fig. 7). While all the other thirty members of showed least algal diversity with one species each (Table 2). Cryptogamic vascular plants viz. *Adiantum caudatum*, *Drynaria quadrifolia* and *Pteris vitata* also had least algal diversity. Three members of the family Malvaceae

(*Sida acuta*, *Hibiscus rosa-sinensis* and *Malva viscus*) and three members of the order Zingiberales (*Kaempferia galanga*, *Maranta aurundinacea* and *Costus speciosus*) had never showed any algal association.

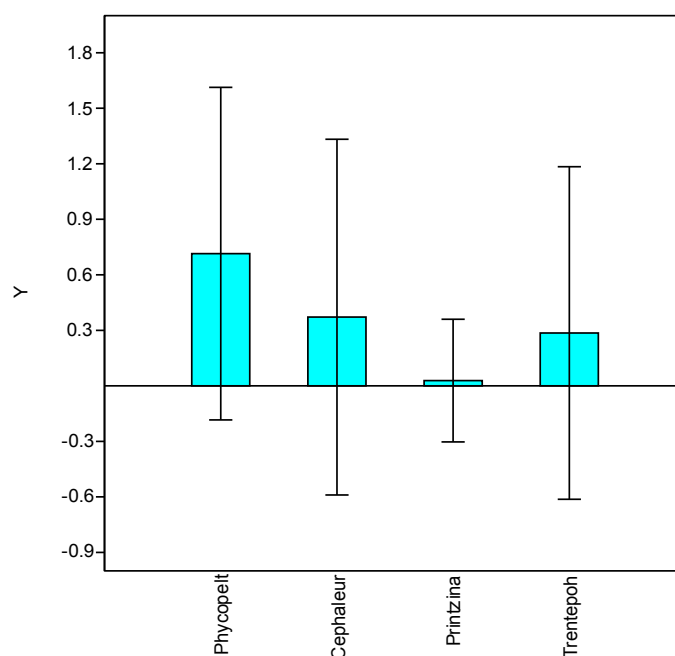


Fig. 4. Bar Chart showing diversity of algae with standard deviation
Abbreviations: Phycopelt- Phycopeltis, Cephalaur-Cephaleuros, Trentepoh-Trentepohlia

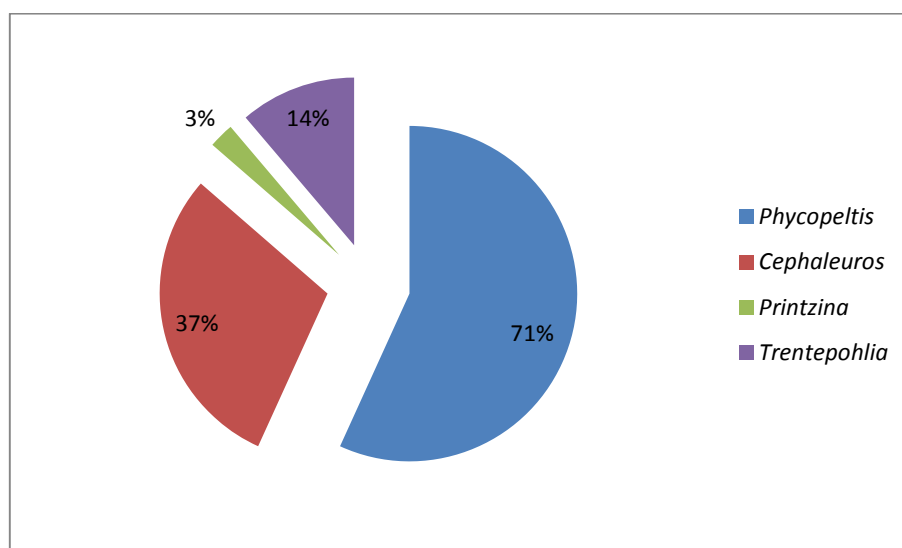


Fig. 5. Pie Chart showing diversity of algae

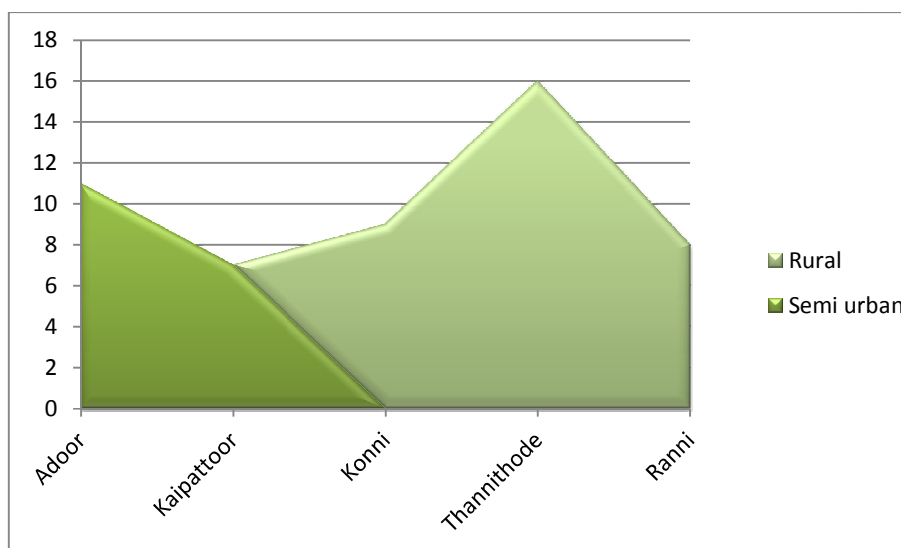


Fig. 6. Species diversity in the area of study

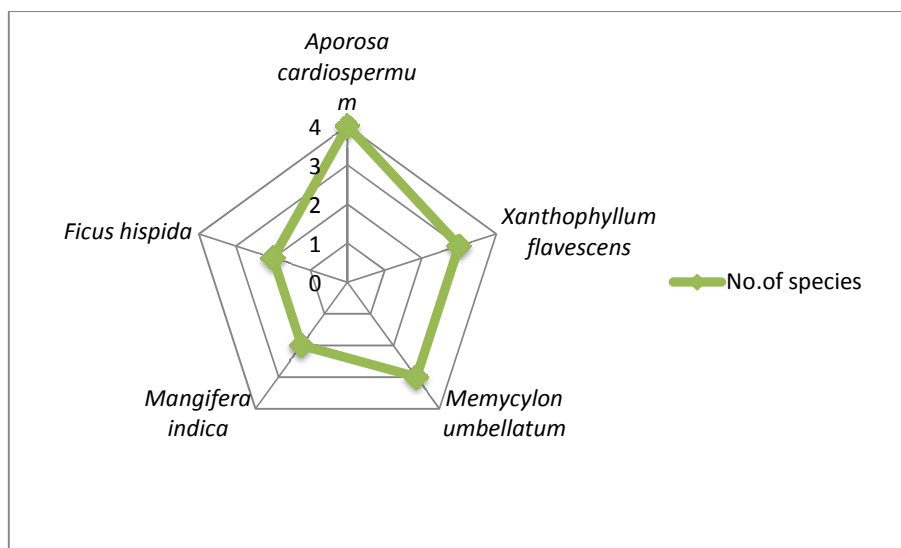


Fig. 7. Trentepohlian algal distribution in various host plants

4. DISCUSSION

The algal flora coming under Trentepohliales are adapted for aerophytic and terrestrial mode of life. They adapted to live either as epiphyllous or endophytic mode of life [16]. Species of Trentepohliales are well known economic and ecological importance except *Cephaleuros*. Trentepohliales are key source of carotenoids and haematochrome [3]. They also serve as the indicators of air pollution. But *Cephaleuros* species are common algal pathogens on the leaves of many vascular plants.

Interestingly it was noted that *Cephaleuros* was seen in the semi-urban areas of agricultural lands [17]. *Printzina* sp. has also exclusively reported from the rural areas of natural forest regions. It may be due to the favorable climatic conditions of the rural areas. *Trentepohlia* spp. was usually found on the bark and leaves of the trees in both semi-urban and rural areas.

Species of Trentepohliales are well flourished not only on the moist rich leaves surfaces but also on the dry bark regions during the South-west and North-east monsoon seasons. The present study revealed that

algae growing on the leaves of natural forest are more diversified than those on semi-urban habitats. The major factors responsible for the distribution of micro flora are moisture, texture, latex and chemical composition of the host plant [18]. Algal flora of the phyllosphere is almost same in tropical and temperate regions. The epiphyllous algal flora observed in the present study was similar to the description [6] from the temperate regions. Maximum diversity of Trentepohliales was observed in the leaves of *Aporosa cardiosperma* followed by *Xanthophyllum flavescens*, *Memecylon umbellatum*, *Mangifera indica* and *Ficus hispida*. On comparing with Phanerogamic plants, Cryptogamic vascular plants had least algal diversity. The members of the family Malvaceae and the order Zingiberales did not show any association of epiphyllous algae.

This is the first report of the diversity of epiphyllous Trentepohlian algal biofilms from the Pathanamthitta District of Kerala. Considering the ecological and economic potentials of Trentepohliales, accounting of its diversity is highly significant.

5. CONCLUSION

Among the four genera of Trentepohliales, *Phycopeltis* spp. showed the maximum abundance in all the seasons. All the genera except *Cephaleuros* exhibit epiphyllous and symbiotic mode of life. But *Cephaleuros* shows endophytic and parasitic mode of life. The rich diversity of Trentepohliales indicates that Pathanamthitta District, an important biodiversity zone of Southern Western Ghats is one of the repositories of epiphyllous algae. Since these species of algae are economically and ecologically important, accounting and exploring its diversity is highly significant.

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COMPETING INTERESTS

Authors have declared that no competing interests exist.

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